# YOLO-R: Road Damage Detector Based on Dynamic Receptive Fields and Lightweight Design

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**DOI:** https://doi.org/10.1007/s11760-025-04523-8

**Abstract:** Automated inspection of road pavement distress is critical for civil infrastructure safety and smart transportation systems. Existing deep learning object detection models frequently struggle with severe background noise, varying crack scales, and limited computing resources on edge patrol platforms. This paper introduces YOLO-R, an advanced lightweight pavement damage detector. Specifically, YOLO-R incorporates dynamic receptive field modeling into the C2f feature extraction module via Scale-Aware Convolution (SAC) embeddings, allowing the receptive field to dynamically adjust to long, thin cracks as well as localized potholes. Furthermore, a GSConv-based slim-neck architecture replaces conventional heavy convolutions, and an ADown module prevents spatial information loss during multi-scale downsampling. Experimental evaluations on the RDD2022 dataset demonstrate that YOLO-R achieves 64.8% mAP@50 while reducing model parameters by 31.4% and FLOPs by 28.6% compared to baseline YOLOv8, enabling real-time edge execution at 84 FPS.

## Key Architectural Contributions

1. **Scale-Aware Dynamic Receptive Fields (SAC in C2f):** Dynamically adapts convolution kernel receptive fields to match high aspect ratio cracks (D00 longitudinal and D10 transverse) versus compact circular potholes (D40).
2. **GSConv Slim-Neck Feature Fusion:** Integrates channel shuffling and depthwise separable convolutions in the neck to preserve semantic correlation with reduced parameter density.
3. **ADown Downsampling Optimization:** Replaces naive strided convolution downsampling with multi-path pooling and point-wise projection, mitigating spatial information loss of subtle road fissures.

## Benchmark Performance Metrics (RDD2022 Dataset)

| Model Architecture | Precision (%) | Recall (%) | mAP@50 (%) | Params (M) | GFLOPs | Inference FPS |
|---|---|---|---|---|---|---|
| YOLOv8n Baseline | 64.9% | 52.6% | 58.5% | 3.0 M | 8.1 G | 92 FPS |
| YOLOv8s Baseline | 68.4% | 54.2% | 60.9% | 11.1 M | 28.4 G | 68 FPS |
| YOLO-R (Proposed) | 70.2% | 57.8% | 64.8% | 4.8 M | 14.2 G | 84 FPS |